Question ID:171
Topic Name: Mathematics-Section A

Question: If $\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right)-\sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right)=\frac{\alpha(60!)}{(30!)(31!)}$, where $\alpha \in \mathbb{R}$, then the value of 16α is equal to

A 1411

B 1320

C 1615

D 1855

Answer Given By Candidate:**Not Attempted**

Question ID:172
Topic Name: Mathematics-Section A

Question: Let a function $f : \mathbb{N} \to \mathbb{N}$ be defined by

$$f(n)=\begin{cases}2n, & n=2,4,6,8,....\\ n-1, & n=3,7,11,15,....\\ \frac{n+1}{2}, & n=1,5,9,13,....\end{cases}$$

then, f is

A one-one but not onto

B onto but not one-one

C neither one-one nor onto

D one-one and onto

Answer Given By Candidate:**D**

Question ID:173

Topic Name: Mathematics-Section A

Question: If the system of linear equations

$$2x + 3y - z = -2$$
$$x + y + z = 4$$
$$x - y + |\lambda|\, z = 4\lambda - 4$$

where $\lambda \in \mathbb{R}$, has no solution, then

A $\lambda = 7$

B $\lambda = -7$

C $\lambda = 8$

D $\lambda^2 = 1$

Answer Given By Candidate:**Not Attempted**

Question ID:174

Topic Name: Mathematics-Section A

Question: Let A be a matrix of order 3×3 and $\det(A) = 2$. Then $\det(\det(A)\ \text{adj}(5\ \text{adj}(A^3)))$ is equal to _____.

A 512×10^6

B 256×10^6

C 1024×10^6

D 256×10^{11}

Answer Given By Candidate:**Not Attempted**

Question ID:175

Topic Name: Mathematics-Section A

Question: The total number of 5-digit numbers, formed by using the digits 1, 2, 3, 5, 6, 7 without repetition, which are multiple of 6, is

A 36

B 48

C 60

D 72

Answer Given By Candidate:**Not Attempted**

Question ID:176
Topic Name: Mathematics-Section A

Question: Let $A_1, A_2, A_3, \ldots$ be an increasing geometric progression of positive real numbers. If $A_1A_3A_5A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then, the value of $A_6 + A_8 + A_{10}$ is equal to

A 33

B 37

C 43

D 47

Answer Given By Candidate:**C**

Question ID:177
Topic Name: Mathematics-Section A

Question: Let $[t]$ denote the greatest integer less than or equal to t. Then, the value of the integral $\int_0^1 [-8x^2 + 6x - 1]\, dx$ is equal to

A -1

B $\frac{-5}{4}$

C $\frac{\sqrt{17} - 13}{8}$

D $\frac{\sqrt{17} - 16}{8}$

Answer Given By Candidate:**Not Attempted**

Question ID:178
Topic Name: Mathematics-Section A

Let $f: \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \le x < 1 \\ b + [\sin(\pi x)], & 1 \le x < 2 \\ [e^{-x}] - c, & x \ge 2 \end{cases}$$

where $a, b, c \in \mathbb{R}$ and $[t]$ denotes greatest integer less than or equal to t. Then, which of the following statements is true ?

Question:

A There exists $a, b, c \in \mathbb{R}$ such that f is continuous on $\mathbb{R}$.

B If f is discontinuous at exactly one point, then $a + b + c = 1$

C If f is discontinuous at exactly one point, then $a + b + c \neq 1$

D f is discontinuous at atleast two points, for any values of a, b and c

Answer Given By Candidate:**Not Attempted**

Question ID:179
Topic Name: Mathematics-Section A
Question: The area of the region $S = \{(x, y) : y^2 \le 8x, y \ge \sqrt{2}x, x \ge 1\}$ is

A $\frac{13\sqrt{2}}{6}$

B $\frac{11\sqrt{2}}{6}$

C $\frac{5\sqrt{2}}{6}$

D $\frac{19\sqrt{2}}{6}$

Answer Given By Candidate:**B**

Question ID:1710
Topic Name: Mathematics-Section A

Let the solution curve $y = y(x)$ of the differential equation

$$\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]x\frac{dy}{dx}=x+\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]y$$

Question: pass through the points $(1, 0)$ and $(2\alpha, \alpha)$, $\alpha > 0$. Then α is equal to

A $\frac{1}{2}\exp\left(\frac{\pi}{6}+\sqrt{e}-1\right)$

B $\frac{1}{2}\exp\left(\frac{\pi}{3}+e-1\right)$

C $\exp\left(\frac{\pi}{6}+\sqrt{e}+1\right)$

D $2\exp\left(\frac{\pi}{3}+\sqrt{e}-1\right)$

Answer Given By Candidate:**Not Attempted**

Question ID:1711
Topic Name: Mathematics-Section A

Question: Let $y = y(x)$ be the solution of the differential equation $x(1-x^2)\frac{dy}{dx}+\left(3x^2y-y-4x^3\right)=0,\ x>1$, with $y(2) = -2$. Then $y(3)$ is equal to

A -18

B -12

C -6

D -3

Answer Given By Candidate:**Not Attempted**

Question ID:1712
Topic Name: Mathematics-Section A

Question: The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____.

A 0

B 1

C 3

D 5

Answer Given By Candidate:**Not Attempted**

Question ID:1713
Topic Name: Mathematics-Section A

Question: Let the eccentricity of the hyperbola $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\sqrt{\frac{5}{2}}$ and length of its latus rectum be $6\sqrt{2}$, If $y = 2x + c$ is a tangent to the hyperbola H, then the value of c^2 is equal to

A 18

B 20

C 24

D 32

Answer Given By Candidate:**Not Attempted**

Question ID:1714
Topic Name: Mathematics-Section A

Question: If the tangents drawn at the points O(0, 0) and $P\left(1+\sqrt{5}, 2\right)$ on the circle $x^2 + y^2 - 2x - 4y = 0$ intersect at the point Q, then the area of the triangle OPQ is equal to

A $\frac{3+\sqrt{5}}{2}$

B $\frac{4+2\sqrt{5}}{2}$

C $\frac{5+3\sqrt{5}}{2}$

D $\frac{7+3\sqrt{5}}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:1715

Topic Name: Mathematics-Section A

Question: If two distinct points Q, R lie on the line of intersection of the planes $-x + 2y - z = 0$ and $3x - 5y + 2z = 0$ and $PQ = PR = \sqrt{18}$ where the point P is $(1, -2, 3)$, then the area of the triangle PQR is equal to

A $\frac{2}{3}\sqrt{38}$

B $\frac{4}{3}\sqrt{38}$

C $\frac{8}{3}\sqrt{38}$

D $\sqrt{\frac{152}{3}}$

Answer Given By Candidate:**Not Attempted**

Question ID:1716

Topic Name: Mathematics-Section A

Question: The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points $(2, 1, 3)$ and $(0, 1, 2)$, respectively, is

A $\frac{\pi}{3}$

B $\frac{\pi}{4}$

C $\frac{\pi}{6}$

D $\frac{\pi}{12}$

Answer Given By Candidate:**Not Attempted**

Question ID:1717

Topic Name: Mathematics-Section A

Question: Let the plane $P: \vec{r} \cdot \vec{a} = d$ contain the line of intersection of two planes $\vec{r} \cdot \left(\hat{i} + 3\hat{j} - \hat{k} \right) = 6$ and $\vec{r} \cdot \left(-6\hat{i} + 5\hat{j} - \hat{k} \right) = 7$. If the plane P passes through the point $\left(2, 3, \frac{1}{2} \right)$, then the value of $\frac{|13\vec{a}|^2}{d^2}$ is equal to

A 90

B 93

C 95

D 97

Answer Given By Candidate:**Not Attempted**

Question ID:1718

Topic Name: Mathematics-Section A

Question: The probability, that in a randomly selected 3-digit number at least two digits are odd, is

A $\frac{19}{36}$

B $\frac{15}{36}$

C $\frac{13}{36}$

D $\frac{23}{36}$

Answer Given By Candidate:**Not Attempted**

Question ID:1719

Topic Name: Mathematics-Section A

Question: Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to

A

$\frac{3-2\sqrt{2}}{2}$

B $\frac{3+\sqrt{2}}{2}$

C $\frac{3-2\sqrt{2}}{4}$

D $\frac{3-\sqrt{2}}{4}$

Answer Given By Candidate:**Not Attempted**

Question ID:1720
Topic Name: Mathematics-Section A

Let p, q, r be three logical statements. Consider the compound statements
$S_1 : ((\sim p) \vee q) \vee ((\sim p) \vee r)$ and
$S_2 : p \rightarrow (q \vee r)$

Question: Then, which of the following is **NOT** true ?

A If S_2 is True, then S_1 is True

B If S_2 is False, then S_1 is False

C If S_2 is False, then S_1 is True

D If S_1 is False, then S_2 is False

Answer Given By Candidate:**C**

Question ID:1721
Topic Name: Mathematics-Section B

Let R_1 and R_2 be relations on the set $\{1, 2, \ldots, 50\}$ such that
$R_1 = \{(p, p^n) : p$ is a prime and $n \geq 0$ is an integer$\}$ and
$R_2 = \{(p, p^n) : p$ is a prime and $n = 0$ or $1\}$.

Question: Then, the number of elements in $R_1 - R_2$ is_____.

Answer Given By Candidate:**Not Attempted**

Question ID:1722
Topic Name: Mathematics-Section B

Question: The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1723
Topic Name: Mathematics-Section B

Question: The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1724
Topic Name: Mathematics-Section B

Question: If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ are coplanar vectors and $\vec{a} \cdot \vec{c} = 5$, $\vec{b} \perp \vec{c}$, then $122\,(c_1 + c_2 + c_3)$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1725
Topic Name: Mathematics-Section B

Question: A ray of light passing through the point P (2, 3) reflects on the x-axis at point A and the reflected ray passes through the point Q (5, 4). Let R be the point that divides the line segment AQ internally into the ratio 2:1. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be (α, β). Then, the value of $7\alpha + 3\beta$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1726
Topic Name: Mathematics-Section B

Question: Let l be a line which is normal to the curve $y = 2x^2 + x + 2$ at a point P on the curve. If the point Q(6, 4) lies on the line l and O is origin, then the area of the triangle OPQ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1727
Topic Name: Mathematics-Section B

Let $A=\{1, a_1, a_2 ..... a_{18}, 77\}$ be a set of integers with $1<a_1<a_2<.....<a_{18}<77$. Let the set $A+A=\{x+y : x, y \in A\}$ contain exactly 39 elements. Then, the value of $a_1+a_2+....+a_{18}$ is equal to

Question: _____.

Answer Given By Candidate:**702**

Question ID:1728

Topic Name: Mathematics-Section B

Question: The number of positive integers k such that the constant term in the binomial expansion of $\left(2x^3+\frac{3}{x^k}\right)^{12}$, $x \neq 0$ is $2^8 \cdot \ell$, where ℓ is an odd integer, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1729

Topic Name: Mathematics-Section B

Question: The number of elements in the set $\{z=a+ib \in \mathbb{C} : a, b \in \mathbb{Z}$ and $1<|z-3+2i|<4\}$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1730

Topic Name: Mathematics-Section B

Question: Let the lines $y+2x=\sqrt{11}+7\sqrt{7}$ and $2y+x=2\sqrt{11}+6\sqrt{7}$ be normal to a circle $C:(x-h)^2+(y-k)^2=r^2$. If the line $\sqrt{11}y-3x=\frac{5\sqrt{77}}{3}+11$ is tangent to the circle C, then the value of $(5h-8k)^2+5r^2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1731

Topic Name: Physics-Section A

Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Product of Pressure (P) and time (t) has the same dimension as that of coeffcient of viscosity.

Reason R : Coefficient of viscosity = $\frac{\text{Force}}{\text{Velocity gradient}}$

Question: Choose the correct answer from the options given below :

A Both **A** and **R** true, and **R** is correct explanation of **A**.

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate:**D**

Question ID:1732

Topic Name: Physics-Section A

Question: A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $a = k^2rt^2$, where k is a constant. The power delivered to the particle by the force acting on it is given as

A zero

B $mk^2r^2t^2$

C mk^2r^2t

D mk^2rt

Answer Given By Candidate:**Not Attempted**

Question ID:1733

Topic Name: Physics-Section A

Question: Motion of a particle in $x-y$ plane is described by a set of following equations $x = 4\sin\left(\frac{\pi}{2} - \omega t\right)$ m and $y = 4\sin(\omega t)$ m. The path of the particle will be :

A circular

B helical

C parabolic

D elliptical

Answer Given By Candidate:**A**

Question ID:1734

Topic Name: Physics-Section A

Match List-I with List-II

	List - I		List - II
A.	Moment of inertia of solid sphere of radius R about any tangent.	I.	$\frac{5}{3}MR^2$
B.	Moment of inertia of hollow sphere of radius (R) about any tangent.	II.	$\frac{7}{5}MR^2$
C.	Moment of inertia of circular ring of radius (R) about its diameter.	III.	$\frac{1}{4}MR^2$
D.	Moment of inertia of circular disc of radius (R) about any diameter.	IV.	$\frac{1}{2}MR^2$

Question: Choose the correct answer from the options given below :

A A-II, B-I, C-IV, D-III

B A-I, B-II, C-IV, D-III

C A-II, B-I, C-III, D-IV

D A-I, B-II, C-III, D-IV

Answer Given By Candidate:**D**

Question ID:1735

Topic Name: Physics-Section A

Question: Two planets A and B of equal mass are having their period of revolutions T_A and T_B such that $T_A = 2T_B$. These planets are revolving in the circular orbits of radii r_A and r_B respectively. Which out of the following would be the correct relationship of their orbits?

A $2r_A^2 = r_B^3$

B $r_A^3 = 2r_B^3$

C $r_A^3 = 4r_B^3$

D $T_A^2 - T_B^2 = \frac{\pi^2}{GM}\left(r_B^3 - 4r_A^3\right)$

Answer Given By Candidate:**C**

Question ID:1736
Topic Name: Physics-Section A

Question: A water drop of diameter 2 cm is broken into 64 equal droplets. The surface tension of water is 0.075 N/m. In this process the gain in surface energy will be :

A 2.8×10^{-4} J

B 1.5×10^{-3} J

C 1.9×10^{-4} J

D 9.4×10^{-5} J

Answer Given By Candidate:**Not Attempted**

Question ID:1737
Topic Name: Physics-Section A

Given below are two statements :
Statement – I : When μ amount of an ideal gas undergoes adiabatic change from state (P_1, V_1, T_1) to state (P_2, V_2, T_2), then work done is $W = \frac{\mu R (T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_p}{C_V}$ and R = universal gas constant.
Statement – II : In the above case, when work is done on the gas, the temperature of the gas would rise.

Question: Choose the correct answer from the options given below:

A Both statement-I and statement-II are true.

B Both statement-I and statement-II are false.

C Statement-I is true but statement-II is false.

D Statement-I is false but statement-II is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1738
Topic Name: Physics-Section A

Given below are two statements :
Statement-I : A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.
Statement-II : An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Question: Choose the correct answer from the options given below :

A Both statement-I and statement-II are true.

B Both statement-I and statement-II are false.

C Statement-I is true but statement-II is false.

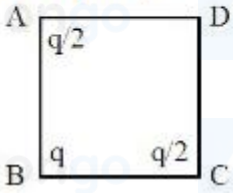
D Statement-I is false but statement-II is true.

Answer Given By Candidate:**A**

Question ID:1739
Topic Name: Physics-Section A

The three charges q/2, q and q/2 are placed at the corners A, B and C of a square of side 'a' as shown in figure. The magnitude of electric field (E) at the corner D of the square, is :

Question:



A $\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

B $\frac{q}{4\pi\epsilon_0 a^2}\left(1+\frac{1}{\sqrt{2}}\right)$

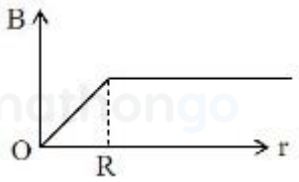
C $\frac{q}{4\pi\epsilon_0 a^2}\left(1-\frac{1}{\sqrt{2}}\right)$

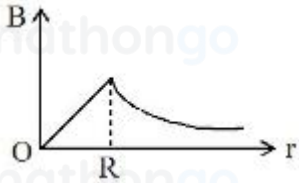
D $\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$

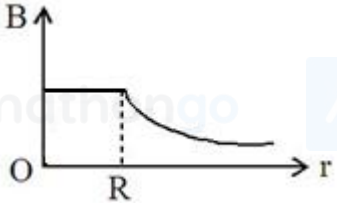
Answer Given By Candidate:**A**

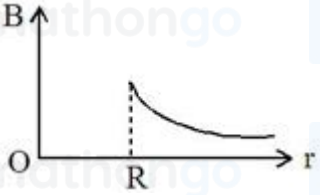
Question ID:1740
Topic Name: Physics-Section A

Question: An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.
Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.

A


B


C


D


Answer Given By Candidate:**D**

Question ID:1741
Topic Name: Physics-Section A

Question: A radar sends an electromagnetic signal of electric field (E_o) = 2.25 V/m and magnetic field (B_o) = 1.5×10^{-8} T which strikes a target on line of sight at a distance of 3 km in a medium. After that, a part of signal (echo) reflects back towards the radar with same velocity and by same path. If the signal was transmitted at time t=0 from radar, then after how much time echo will reach to the radar?

A 2.0×10^{-5} s

B 4.0×10^{-5} s

C 1.0×10^{-5} s

D 8.0×10^{-5} s

Answer Given By Candidate:**B**

Question ID:1742

Topic Name: Physics-Section A

Question: The refracting angle of a prism is A and refractive index of the material of the prism is cot (A/2). Then the angle of minimum deviation will be -

A $180 - 2A$

B $90 - A$

C $180 + 2A$

D $180 - 3A$

Answer Given By Candidate:**Not Attempted**

Question ID:1743

Topic Name: Physics-Section A

Question: The aperture of the objective is 24.4 cm. The resolving power of this telescope, if a light of wavelength 2440 $\mathring{A}$ is used to see the object will be :

A 8.1×10^{6}

B 10.0×10^{7}

C 8.2×10^{5}

D 1.0×10^{-8}

Answer Given By Candidate:**Not Attempted**

Question ID:1744

Topic Name: Physics-Section A

Question: The de Broglie wavelengths for an electron and a photon are λ_e and λ_p respectively. For the same kinetic energy of electron and photon, which of the following presents the correct relation between the de Broglie wavelengths of two ?

A $\lambda_p \propto \lambda_e^2$

B $\lambda_p \propto \lambda_e$

C $\lambda_p \propto \sqrt{\lambda_e}$

D $\lambda_p \propto \sqrt{\frac{1}{\lambda_e}}$

Answer Given By Candidate:**B**

Question ID:1745

Topic Name: Physics-Section A

Question: The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as :

A $Q = K_p$

B $(K_p + Q) < 0$

C $Q < K_p$

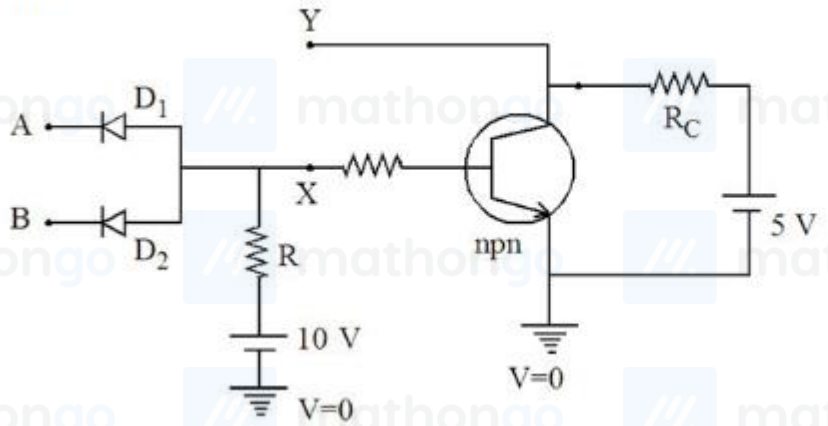
D $(K_p + Q) > 0$

Answer Given By Candidate:**Not Attempted**

Question ID:1746

Topic Name: Physics-Section A

Question: In the following circuit, the correct relation between output (Y) and inputs A and B will be :



A $Y = AB$

B $Y = A + B$

C $Y = \overline{AB}$

D $Y = \overline{A + B}$

Answer Given By Candidate:**Not Attempted**

Question ID:1747
Topic Name: Physics-Section A

Question: For using a multimeter to identify diode from electrical components, choose the correct statement out of the following about the diode :

A It is two terminal device which conducts current in both directions.

B It is two terminal device which conducts current in one direction only

C It does not conduct current gives an initial deflection which decays to zero.

D It is three terminal device which conducts current in one direction only between central terminal and either of the remaining two terminals

Answer Given By Candidate:**Not Attempted**

Question ID:1748
Topic Name: Physics-Section A

Question: Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A : n-p-n transistor permits more current than a p-n-p transistor.
Reason R : Electrons have greater mobility as a charge carrier.
Choose the correct answer from the options given below :

A Both **A** and **R** true, and **R** is correct explanation of **A**.

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1749
Topic Name: Physics-Section A

Question: Match List-I with List-II

	List - I		List - II
A.	Television signal	I.	03 KHz
B.	Radio signal	II.	20 KHz
C.	High Quality Music	III.	02 MHz
D.	Human speech	IV.	06 MHz

Choose the correct answer from the options given below :

A A-I, B-II, C-III, D-IV

B A-IV, B-III, C-I, D-II

C A-IV, B-III, C-II, D-I

D A-I, B-II, C-IV, D-III

Answer Given By Candidate:**D**

Question ID:1750

Topic Name: Physics-Section A

Question: The velocity of sound in a gas, in which two wavelengths 4.08m and 4.16m produce 40 beats in 12s, will be :

A 282.8 ms^{-1}

B 175.5 ms^{-1}

C 353.6 ms^{-1}

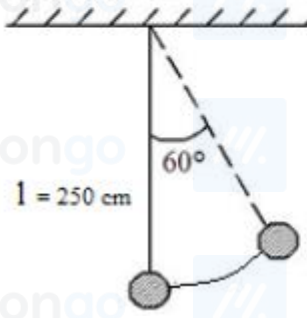
D 707.2 ms^{-1}

Answer Given By Candidate:**Not Attempted**

Question ID:1751

Topic Name: Physics-Section B

Question: A pendulum is suspended by a string of length 250 cm. The mass of the bob of the pendulum is 200 g. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob, the maximum velocity attained by the bob will be _____ ms^{-1}. (if g = 10 m/s^2)

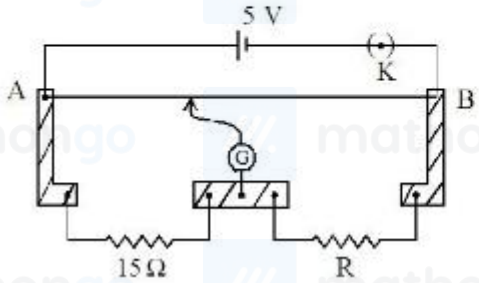


Answer Given By Candidate:**Not Attempted**

Question ID:1752

Topic Name: Physics-Section B

A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15 Ω. The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A. If the end correction for end A is 2 cm, then the determined value of R will be _____ Ω.

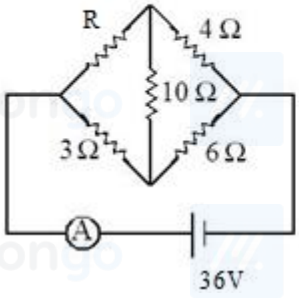


Question:

Answer Given By Candidate:**18**

Question ID:1753
Topic Name: Physics-Section B

Current measured by the ammeter Ⓐ in the reported circuit when no current flows through 10 Ω resistance, will be _____ A.

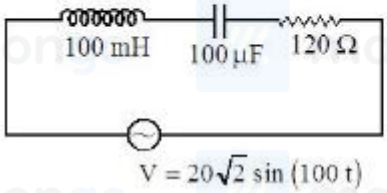


Question:

Answer Given By Candidate:**4**

Question ID:1754
Topic Name: Physics-Section B

An AC source is connected to an inductance of 100 mH, a capacitance of 100 μ F and a resistance of 120 Ω as shown in figure. The time in which the resistance having a thermal capacity 2 J/°C will get heated by 16°C is _____ s.



Question:

Answer Given By Candidate:**Not Attempted**

Question ID:1755
Topic Name: Physics-Section B

Question: The position vector of 1 kg object is $\vec{r} = \left(3\hat{i} - \hat{j}\right)$ m and its velocity $\vec{v} = \left(3\hat{j} + \hat{k}\right) ms^{-1}$. The magnitude of its angular momentum is $\sqrt{x}$ Nm where x is _____.

Answer Given By Candidate:**91**

Question ID:1756

Topic Name: Physics-Section B

Question: A man of 60 kg is running on the road and suddenly jumps into a stationary trolly car of mass 120 kg. Then, the trolly car starts moving with velocity 2 ms^{-1}. The velocity of the running man was _____ ms^{-1}, when he jumps into the car.

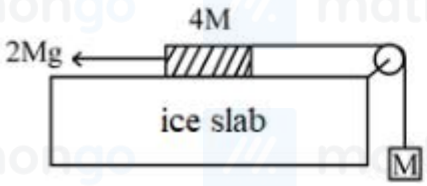
Answer Given By Candidate:**6**

Question ID:1757

Topic Name: Physics-Section B

Question: A hanging mass M is connected to a four times bigger mass by using a string-pulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation, tension in the string is $\frac{x}{5}$ Mg for x = _____. Neglect mass of the string and friction of the block (bigger mass) with ice slab.

(Given g = acceleration due to gravity)



Answer Given By Candidate:**5**

Question ID:1758

Topic Name: Physics-Section B

Question: The total internal energy of two mole monoatomic ideal gas at temperature T = 300 K will be _____J. (Given R = 8.31 J/mol.K)

Answer Given By Candidate:**Not Attempted**

Question ID:1759

Topic Name: Physics-Section B

Question: A singly ionized magnesium atom (A=24) ion is accelerated to kinetic energy 5 keV, and is projected perpendicularly into a magnetic field B of the magnitude 0.5 T. The radius of path formed will be _____ cm.

Answer Given By Candidate:**Not Attempted**

Question ID:1760

Topic Name: Physics-Section B

Question: A telegraph line of length 100 km has a capacity of 0.01 μF/km and it carries an alternating current at 0.5 kilo cycle per second. If minimum impedance is required, then the value of the inductance that needs to be introduced in series is _____ mH. (if $\pi = \sqrt{10}$)

Answer Given By Candidate:**Not Attempted**

Question ID:1761

Topic Name: Chemistry-Section A

Question: The **incorrect** statement about the imperfections in solids is :

A Schottky defect decreases the density of the substance.

B Interstitial defect increases the density of the substance.

C Frenkel defect does not alter the density of the substance.

D Vacancy defect increases the density of the substance.

Answer Given By Candidate:**D**

Question ID:1762

Topic Name: Chemistry-Section A

Question: The Zeta potential is related to which property of colloids ?

A Colour

B Tyndall effect

C Charge on the surface of colloidal particles

D Brownian movement

Answer Given By Candidate:**Not Attempted**

Question ID:1763

Topic Name: Chemistry-Section A

Question: Element "E" belongs to the period 4 and group 16 of the periodic table. The valence shell electron configuration of the element, which is just above "E" in the group is

A $3s^2, 3p^4$

B $3d^{10}, 4s^2, 4p^4$

C $4d^{10}, 5s^2, 5p^4$

D $2s^2, 2p^4$

Answer Given By Candidate:**B**

Question ID:1764

Topic Name: Chemistry-Section A

Question: Given are two statements one is labelled as **Assertion A** and other is labelled as **Reason R**.
Assertion A :Magnesium can reduce Al_2O_3 at a temperature below 1350°C, while above 1350°C aluminium can reduce MgO.
Reason R : The melting and boiling points of magnesium are lower than those of aluminium.

In light of the above statements, choose most appropriate answer from the options given below :

A Both **A** and **R** are correct, and **R** is correct explanation of A.

B Both **A** and **R** are correct, but **R** is NOT the correct explanation of **A**.

C **A** is correct **R** is not correct.

D **A** is not correct, **R** is correct.

Answer Given By Candidate:**Not Attempted**

Question ID:1765

Topic Name: Chemistry-Section A

Question: Dihydrogen reacts with CuO to give

A CuH_2

B Cu

C Cu_2O

D $Cu(OH)_2$

Answer Given By Candidate:**A**

Question ID:1766
Topic Name: Chemistry-Section A
Question: Nitrogen gas is obtained by thermal decomposition of

A $Ba(NO_3)_2$

B $Ba(N_3)_2$

C $NaNO_2$

D $NaNO_3$

Answer Given By Candidate:**C**

Question ID:1767
Topic Name: Chemistry-Section A
Question: Given below are two statements :
Statement **I** :The pentavalent oxide of group-15 element, E_2O_5, is less acidic than trivalent oxide, E_2O_3, of the same element.
Statement **II** :The acidic character of trivalent oxide of group 15 elements, E_2O_3, decreases down the group.

In light of the above statements, choose **most appropriate** answer from the options given below :

A Both Statement I and Statement II are true.

B Both Statement I and Statement II are false.

C Statement I true, but statement II is false.

D Statement I is false but statement II is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1768
Topic Name: Chemistry-Section A
Question: Which one of the lanthanoids given below is the most stable in divalent form?

A Ce (Atomic Number 58)

B Sm (Atomic Number 62)

C Eu (Atomic Number 63)

D Yb (Atomic Number 70)

Answer Given By Candidate:**Not Attempted**

Question ID:1769

Topic Name: Chemistry-Section A

Question: Given below are two statements :

Statement **I** : $[Ni(CN)_4]^{2-}$ is square planar and diamagnetic complex, with dsp^2 hybridization for Ni but $[Ni(CO)_4]$ is tetrahedral, paramagnetic and with sp^3-hybridication for Ni.

Statement **II** : $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ both have same d-electron configuration, have same geometry and are paramagnetic.

In light the above statements, choose the **correct** answer form the options given below :

A Both Statement I and Statement II are true.

B Both Statement I and Statement II are false.

C Statement I is correct but statement II is false.

D Statement I is incorrect but statement II is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1770

Topic Name: Chemistry-Section A

Question: Which amongst the following is **not** a pesticide ?

A DDT

B Organophosphates

C Dieldrin

D Sodium arsenite

Answer Given By Candidate:**Not Attempted**

Question ID:1771

Topic Name: Chemistry-Section A

Question: Which one of the following techniques is **not** used to spot components of a mixture separated on thin layer chromatographic plate ?

A I_2 (Solid)

B U.V. Light

C Visualisation agent as a component of mobile phase

D Spraying of an appropriate reagent

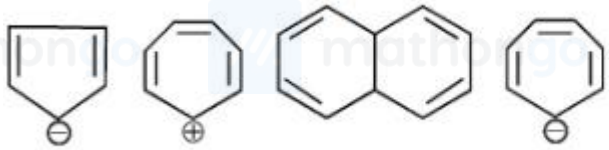
Answer Given By Candidate:**Not Attempted**

Question ID:1772

Topic Name: Chemistry-Section A

Which of the following structures are aromatic in nature ?

Question:



A A, B, C and D

B Only A and B

C Only A and C

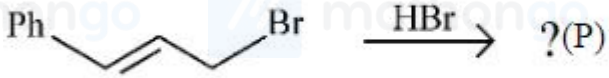
D Only B, C and D

Answer Given By Candidate:**B**

Question ID:1773

Topic Name: Chemistry-Section A

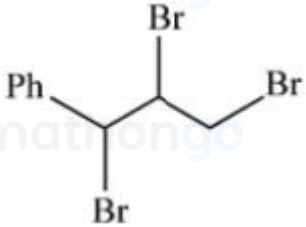
The major product (P) in the reaction



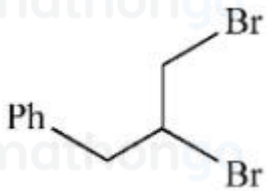
[Ph is - C_6H_5]

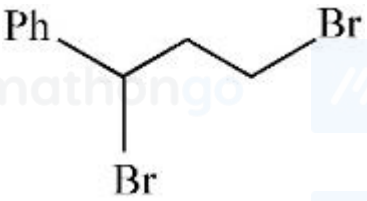
Question: is

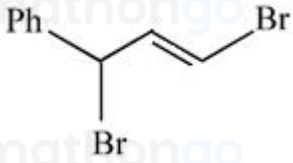
A



B



C 

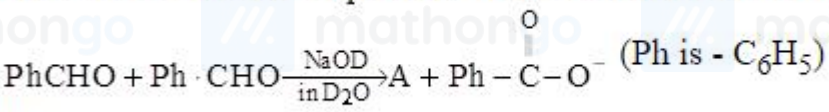
D 

Answer Given By Candidate:**C**

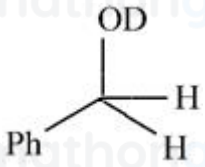
Question ID:1774

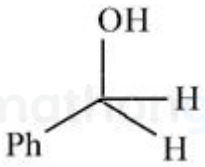
Topic Name: Chemistry-Section A

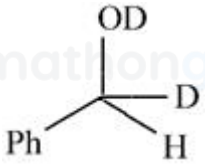
The correct structure of product 'A' formed in the following reaction,

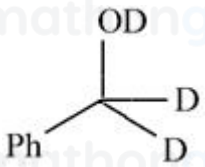


Question: is

A 

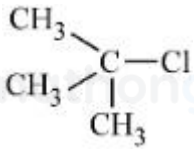
B 

C 

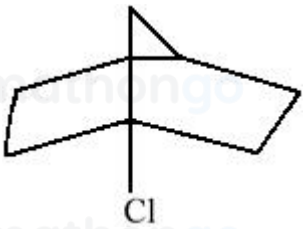
D 

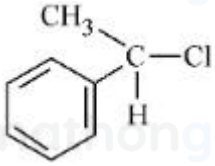
Answer Given By Candidate:**Not Attempted**

Question ID:1775
Topic Name: Chemistry-Section A
Question: Which one of the following compounds is inactive towards S_N1 reaction ?

A 

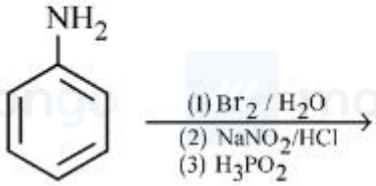
B 

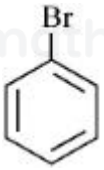
C 

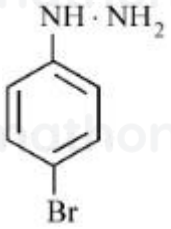
D 

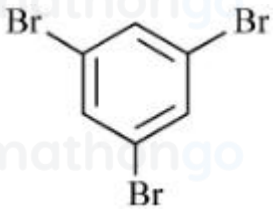
Answer Given By Candidate:**A**

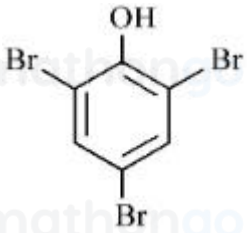
Question ID:1776
Topic Name: Chemistry-Section A
Question: Identify the major product formed in the following sequence of reactions :



A 

B 

C 

D 

Answer Given By Candidate:**Not Attempted**

Question ID:1777
Topic Name: Chemistry-Section A

A primary aliphatic amine on reaction with nitrous acid in cold (273 K) and there after raising temperature of reaction mixture to room temperature (298 K), gives

Question: a/an

A nitrile

B alcohol

C diazonium salt

D secondary amine

Answer Given By Candidate:**Not Attempted**

Question ID:1778
Topic Name: Chemistry-Section A
Question: Which one of the following is **NOT** a copolymer ?

A Buna-S

B Neoprene

C PHBV

D Butadiene-styrene

Answer Given By Candidate:**Not Attempted**

Question ID:1779
Topic Name: Chemistry-Section A

Question: Stability of α - Helix structure of proteins depends upon

A dipolar interaction

B H-bonding interaction

C van der Waals forces

D π -stacking interaction

Answer Given By Candidate: **B**

Question ID: 1780

Topic Name: Chemistry-Section A

Question: The formula of the purple colour formed in Laissaigne's test for sulphur using sodium nitroprusside is

A $NaFe[Fe(CN)_6]$

B $Na[Cr(NH_3)_2(NCS)_4]$

C $Na_2[Fe(CN)_5(NO)]$

D $Na_4[Fe(CN)_5(NOS)]$

Answer Given By Candidate: **Not Attempted**

Question ID: 1781

Topic Name: Chemistry-Section B

Question: A 2.0 g sample containing MnO_2 is treated with HCl liberating Cl_2. The Cl_2 gas is passed into a solution of KI and 60.0 mL of 0.1 M $Na_2S_2O_3$ is required to titrate the liberated iodine. The percentage of MnO_2 in the sample is _____. (Nearest integer)
[Atomic masses (in u) Mn = 55; Cl = 35.5; O = 16, I = 127, Na = 23, K = 39, S = 32]

Answer Given By Candidate: **Not Attempted**

Question ID: 1782

Topic Name: Chemistry-Section B

Question: If the work function of a metal is 6.63×10^{-19} J, the maximum wavelength of the photon required to remove a photoelectron from the metal is _____ nm. (Nearest integer)
[Given : $h = 6.63 \times 10^{-34}$ J s, and $c = 3 \times 10^8$ m s^{-1}]

Answer Given By Candidate:**300**

Question ID:1783
Topic Name: Chemistry-Section B
Question: The hybridization of P exhibited in PF_5 is sp^xd^y. The value of y is _____

Answer Given By Candidate:**2**

Question ID:1784
Topic Name: Chemistry-Section B
Question: 4.0 L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 20 L. The amount of heat absorbed in this expansion is _____ L atm.

Answer Given By Candidate:**Not Attempted**

Question ID:1785
Topic Name: Chemistry-Section B
Question: The vapour pressures of two volatile liquids A and B at 25°C are 50 Torr and 100 Torr, respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is _____.

Answer Given By Candidate:**10**

Question ID:1786
Topic Name: Chemistry-Section B
Question: The solubility product of a sparingly soluble salt A_2X_3 is 1.1×10^{-23}. If specific conductance of the solution is 3×10^{-5} S m^{-1}, the limiting molar conductivity of the solution is $x \times 10^{-3}$ S m^2 mol^{-1}. The value of x is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1787
Topic Name: Chemistry-Section B
Question: The quantity of electricity in Faraday needed to reduce 1 mol of $Cr_2O_7^{2-}$ to Cr^{3+} is _____

Answer Given By Candidate:**Not Attempted**

Question ID:1788
Topic Name: Chemistry-Section B

Question: For a first order reaction A → B, the rate constant, $k = 5.5 \times 10^{-14}\ s^{-1}$. The time required for 67% completion of reaction is $x \times 10^{-1}$ times the half life of reaction. The value of x is _____ (Nearest integer)
(Given : log 3 = 0.4771)

Answer Given By Candidate:**Not Attempted**

Question ID:1789
Topic Name: Chemistry-Section B

Question: Number of complexes which will exhibit synergic bonding amongst, $[Cr(CO)_6]$, $[Mn(CO)_5]$ and $[Mn_2(CO)_{10}]$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1790
Topic Name: Chemistry-Section B

Question: In the estimation of bromine, 0.5 g of an organic compound gave 0.40 g of silver bromide. The percentage of bromine in the given compound is _____% (nearest integer)
(Relative atomic masses of Ag and Br are 108u and 80u, respectively).

Answer Given By Candidate:**Not Attempted**